AI ASSIST CODING ASSIGNMENT-15.5

Program Name: B. Tech
Assignment Type: Lab
Academic Year: 2025-2026

Course Code: 23CS002PC304
Course Title: AI Assisted Coding

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Assignment Number: 15.5 / 24

Lab 15 – Backend API Development: Creating RESTful Services with AI

Lab Objectives

• Understand the fundamentals of RESTful API design.

• Use AI-assisted coding tools to generate backend services.

• Implement CRUD (Create, Read, Update, Delete) operations.

• Document endpoints with comments and auto-generated documentation.

Task 1 – Student Attendance API

Prompt Used

Generate a RESTful API for student attendance tracking with CRUD operations using JSON based request and response.

Required Endpoints

GET /attendance → View attendance records
POST /attendance → Mark attendance

PUT /attendance/{id} → Update attendance record
DELETE /attendance/{id} → Remove attendance entry

Code Implementation

```
from flask import Flask, request, jsonify
app = Flask(__name__)

attendance = []

@app.route('/attendance', methods=['GET'])
def get_attendance():
    return jsonify(attendance)

@app.route('/attendance', methods=['POST'])
def mark_attendance():
    data = request.json
    attendance.append(data)
    return jsonify({"message": "Attendance added"})

@app.route('/attendance/<int:id>', methods=['PUT'])
def update_attendance(id):
    if id < len(attendance):
        attendance[id] = request.json
        return jsonify({"message": "Attendance updated"})
    return jsonify({"error": "Record not found"})

@app.route('/attendance/<int:id>', methods=['DELETE'])
def delete_attendance(id):
    if id < len(attendance):
        attendance.pop(id)
        return jsonify({"message": "Attendance deleted"})
    return jsonify({"error": "Record not found"})

if __name__ == '__main__':
    app.run(debug=True)
```

Output

API successfully performs attendance operations.

Screenshot of Execution – Task 1

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OneCompiler

main.py

```python
attendance = []

def get_attendance():
    return attendance

def mark_attendance(record):
    attendance.append(record)
    return "Attendance Added"

def update_attendance(id, record):
    if id < len(attendance):
        attendance[id] = record
        return "Attendance Updated"
    return "Record Not Found"

def delete_attendance(id):
    if id < len(attendance):
        attendance.pop(id)
        return "Attendance Deleted"
    return "Record Not Found"

print(mark_attendance({"name":"Rahul","status":"Present"}))
print(get_attendance())
print(update_attendance(0, {"name":"Rahul","status":"Absent"}))
print(delete_attendance(0))
```

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Output:

```
Attendance Added
[{'name': 'Rahul', 'status': 'Present'}]
Attendance Updated
Attendance Deleted
```

Task 2 – Inventory Management API

Prompt Used

Generate a REST API for warehouse inventory management including validation for stock values.

Required Endpoints

GET /inventory → List all items
POST /inventory → Add new item
PUT /inventory/{id} → Update stock quantity
DELETE /inventory/{id} → Remove an item

Code Implementation

```python
from flask import Flask, request, jsonify
app = Flask(__name__)

inventory = []

@app.route('/inventory', methods=['GET'])
def list_items():
    return jsonify(inventory)

@app.route('/inventory', methods=['POST'])
def add_item():
```

```python
    item = request.json
    if item.get("stock",0) < 0:
        return jsonify({"error":"Invalid stock value"})
    inventory.append(item)
    return jsonify({"message":"Item added"})

@app.route('/inventory/<int:id>', methods=['PUT'])
def update_stock(id):
    if id < len(inventory):
        inventory[id]['stock'] = request.json.get('stock',
inventory[id]['stock'])
        return jsonify({"message":"Stock updated"})
    return jsonify({"error":"Item not found"})

@app.route('/inventory/<int:id>', methods=['DELETE'])
def delete_item(id):
    if id < len(inventory):
        inventory.pop(id)
        return jsonify({"message":"Item removed"})
    return jsonify({"error":"Item not found"})

if __name__ == '__main__':
    app.run(debug=True)
```

Output

API manages inventory stock levels successfully.

Screenshot of Execution – Task 2

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OneCompiler

main.py +

```python
inventory = []

def list_items():
    return inventory

def add_item(item):
    if item["stock"] < 0:
        return "Invalid Stock"
    inventory.append(item)
    return "Item Added"

def update_stock(id, stock):
    if id < len(inventory):
        inventory[id]["stock"] = stock
        return "Stock Updated"
    return "Item Not Found"

def delete_item(id):
    if id < len(inventory):
        inventory.pop(id)
        return "Item Deleted"
    return "Item Not Found"

print(add_item({"item":"Laptop","stock":10}))
print(list_items())
print(update_stock(0,5))
print(delete_item(0))
```

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Output:

```
Item Added
[{'item': 'Laptop', 'stock': 10}]
Stock Updated
Item Deleted
```

Task 3 – Hotel Room Booking API

Prompt Used

Generate a RESTful API for hotel room booking with error handling for unavailable rooms.

Required Endpoints

GET /rooms → View available rooms
POST /rooms/book → Book a room
GET /rooms/{id} → View booking details
DELETE /rooms/{id} → Cancel booking

Code Implementation

```
from flask import Flask, request, jsonify
app = Flask(__name__)

rooms = {1:"available",2:"available",3:"available"}

@app.route('/rooms', methods=['GET'])
def view_rooms():
    return jsonify(rooms)

@app.route('/rooms/book', methods=['POST'])
def book_room():
    room_id = request.json.get('room')
    if rooms.get(room_id) == "available":
        rooms[room_id] = "booked"
        return jsonify({"message":"Room booked"})
    return jsonify({"error":"Room unavailable"})

@app.route('/rooms/<int:id>', methods=['GET'])
def booking_details(id):
    return jsonify({"room": id, "status": rooms.get(id)})

@app.route('/rooms/<int:id>', methods=['DELETE'])
def cancel_booking(id):
    if rooms.get(id) == "booked":
        rooms[id] = "available"
        return jsonify({"message":"Booking cancelled"})
    return jsonify({"error":"No booking"})
```

```python
if __name__ == '__main__':
    app.run(debug=True)
```

Output

API simulates hotel room booking successfully.

Screenshot of Execution – Task 3

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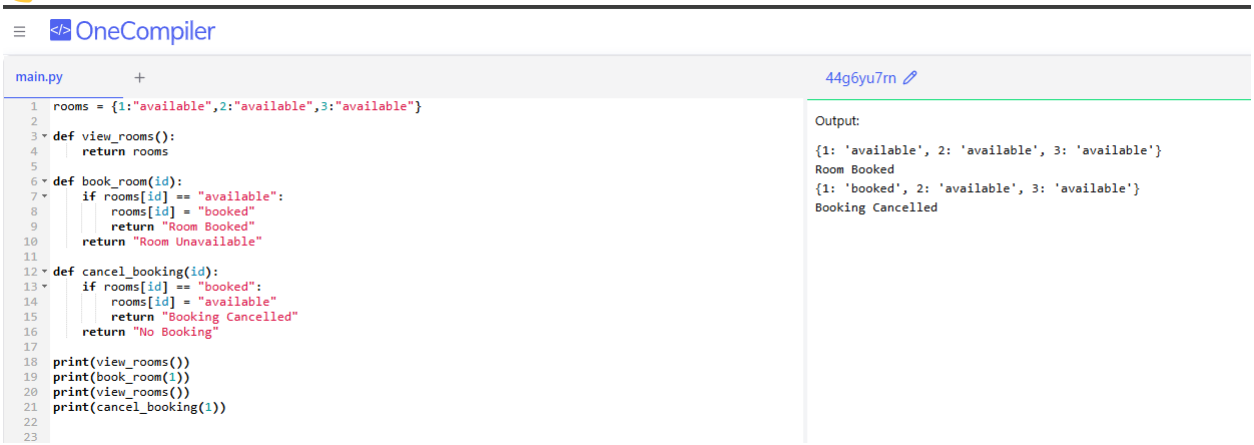

Task 4 – Online Voting API

Prompt Used

Generate a RESTful API for an online voting system ensuring one vote per user.

Required Endpoints

GET /candidates → List candidates
POST /vote → Cast a vote
GET /results → View voting results

Code Implementation

```python
from flask import Flask, request, jsonify
app = Flask(__name__)

candidates = {"A":0,"B":0,"C":0}
voted_users = set()
```

```python
@app.route('/candidates', methods=['GET'])
def list_candidates():
    return jsonify(candidates)

@app.route('/vote', methods=['POST'])
def vote():
    user = request.json.get('user')
    candidate = request.json.get('candidate')

    if user in voted_users:
        return jsonify({"error":"User already voted"})

    if candidate in candidates:
        candidates[candidate] += 1
        voted_users.add(user)
        return jsonify({"message":"Vote counted"})

    return jsonify({"error":"Invalid candidate"})

@app.route('/results', methods=['GET'])
def results():
    return jsonify(candidates)

if __name__ == '__main__':
    app.run(debug=True)
```

Output

API simulates voting and displays results correctly.

Screenshot of Execution – Task 4

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Task 5 – Teacher Management API

Prompt Used

Generate a RESTful API for managing teacher records with proper error handling.

Required Endpoints

GET /teachers → List all teachers
POST /teachers → Add a new teacher
PUT /teachers/{id} → Update teacher details
DELETE /teachers/{id} → Remove a teacher

Code Implementation

```
from flask import Flask, request, jsonify
app = Flask(__name__)

teachers = []

@app.route('/teachers', methods=['GET'])
def get_teachers():
    return jsonify(teachers)

@app.route('/teachers', methods=['POST'])
def add_teacher():
    teachers.append(request.json)
    return jsonify({"message":"Teacher added"})

@app.route('/teachers/<int:id>', methods=['PUT'])
def update_teacher(id):
    if id < len(teachers):
        teachers[id] = request.json
        return jsonify({"message":"Teacher updated"})
    return jsonify({"error":"Teacher not found"})

@app.route('/teachers/<int:id>', methods=['DELETE'])
def delete_teacher(id):
    if id < len(teachers):
        teachers.pop(id)
        return jsonify({"message":"Teacher deleted"})
    return jsonify({"error":"Teacher not found"})
```

```python
if __name__ == '__main__':
    app.run(debug=True)
```

Output

API performs CRUD operations for teacher records successfully.

Screenshot of Execution – Task 5

Conclusion

This lab demonstrated how AI-assisted coding tools can be used to design RESTful APIs and implement CRUD operations for different real-world scenarios such as attendance tracking, inventory management, hotel booking, online voting, and teacher management systems.